

CHARACTERIZATION OF CRYSTAL EFFICIENCY:

STATISTICAL AND SYSTEMATIC ERRORS

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PREVIOUS RESULTS

	τ_y [$\mu\text{rad}/\text{mm}$]	θ_b [μrad]	ϵ_{ch} [%]
Long crystal 1	5.98 ± 0.03	6008.4 ± 0.1	19.4 ± 0.01
Long crystal 2	3.77 ± 0.01	6122.3 ± 0.2	18.3 ± 0.01

UNCERTAINTY EVALUATION

- Statistical uncertainty ($\sim 0.1\%$)
 - Binomial fluctuation of the particle count
 - Fit errors

- Systematic uncertainty ($\sim 1.5\%$):

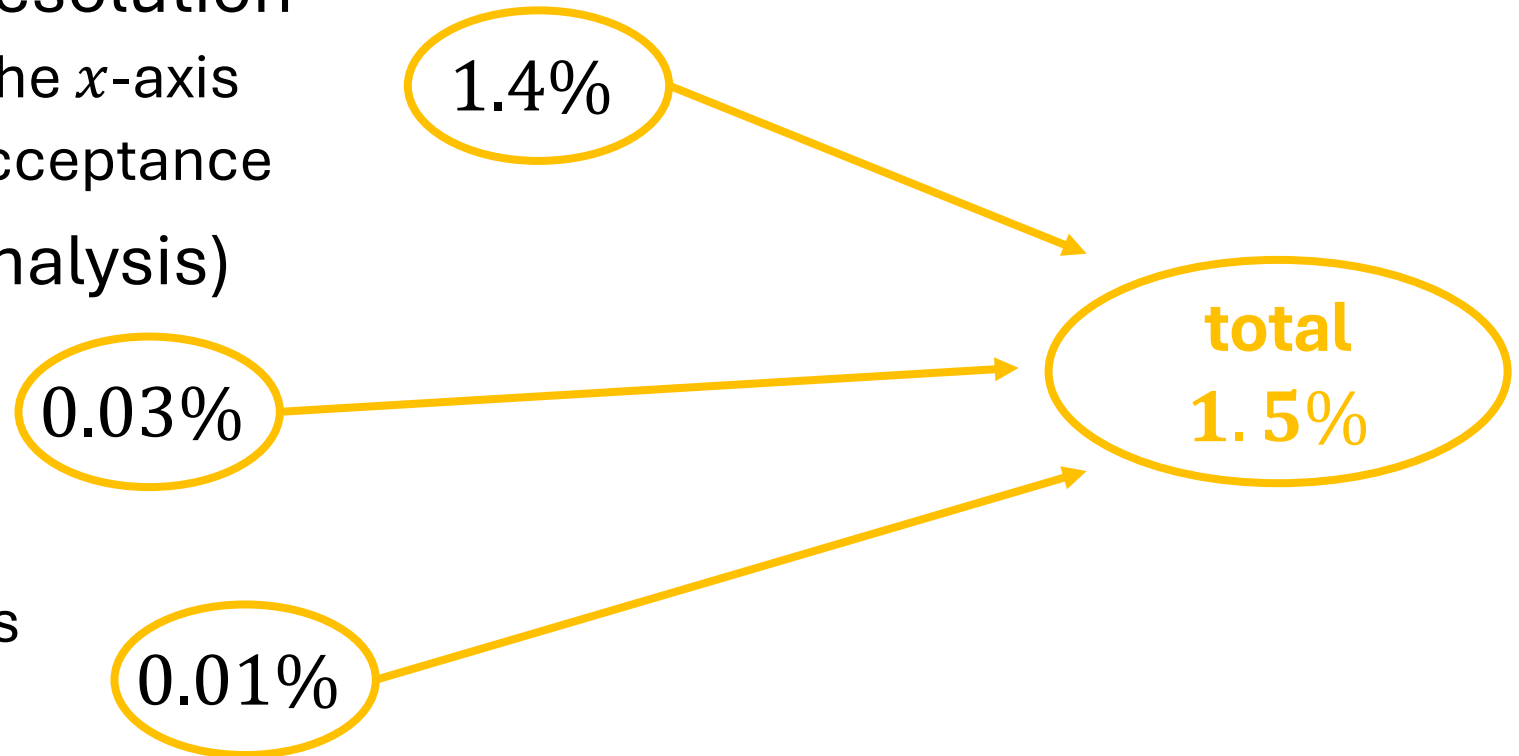
evaluated by individually perturbing the key parameters of the analysis algorithm and computing the half spread of the efficiency:

$$\frac{|\epsilon_{up} - \epsilon_{down}|}{2}$$

SYSTEMATIC ERRORS

Classification of sources:

- Dominant: tracker resolution
 - Spatial cuts along the x -axis
 - Lindhard angular acceptance
- Modeling (torsion analysis)
 - Fit model choice
 - Grid binning
- Selection
 - Integration windows
 - Preliminary cuts



SYSTEMATIC ERRORS: DOMINANT

- Shift of the margins along the bending axis (x): 1.10%
 - The margins of the spatial box were widened and narrowed by a value equal to the position error ~ 0.1 mm
 - The width of the crystal along x is 2 mm $\rightarrow$ 5% of the active face
 - At the margins the statistical distribution decreases, the angular acceptance is less precise (miscut, edge damage ...) so the signal becomes less reliable
 - $\rightarrow$ consistent with the fact that we do not consider them in our analysis
- Variation of the critical angle cut (θ_L): 0.98%
 - The angular resolution of the tracker ($\sim 8.9 \mu\text{rad}$) is broader than the Lindhard half-angle ($\sim 6.5 \mu\text{rad}$).
 - Performing the analysis with a small shift of the acceptance window tested the sensitivity of the N_{ch} count

SYSTEMATIC ERRORS: MODELING

Recalculating the entire filtering chain to prove that the algorithmic choices do not introduce bias:

- Fit model choice: [0.03% – 0.20%]
 - Using different models: parabolic model, quadratic model (which includes mixed terms)
- Binning choice: [0.005%]
 - Variation of the number of slices for the local gaussian scan

The stability of the torsion value and of the efficiency proves the correction does not depend on the statistical fluctuations.

SYSTEMATIC ERRORS: SELECTION

- These systematics test arbitrary “cut-based” choices made ($\mu \pm 3\sigma$).
- The overall impact is well below 0.1% conforming the absence of human bias.

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The final calculated value is robust from a data analysis perspective. The only real limitations seems to come from the telescope resolution, not from the software or the assumptions made to extract the torsion map.

ERROR BUDGET SUMMARY

run	efficiency	error	stat error	syst err (dom)	syst err (model)	syst err (select)
crystal 1	19.6%	1.5%	0.1%	1.5%	0.03%	0.1%
	19.1%	1.5%	0.1%	1.4%	0.2%	0.1%
	17.1%	1.3%	0.1%	1.3%	0.05%	0.07%
crystal 2	18.2%	1.4%	0.1%	1.4%	0.2%	0.2%
	18.3%	1.5%	0.1%	1.4%	0.1%	0.2%

ERROR BUDGET SUMMARY

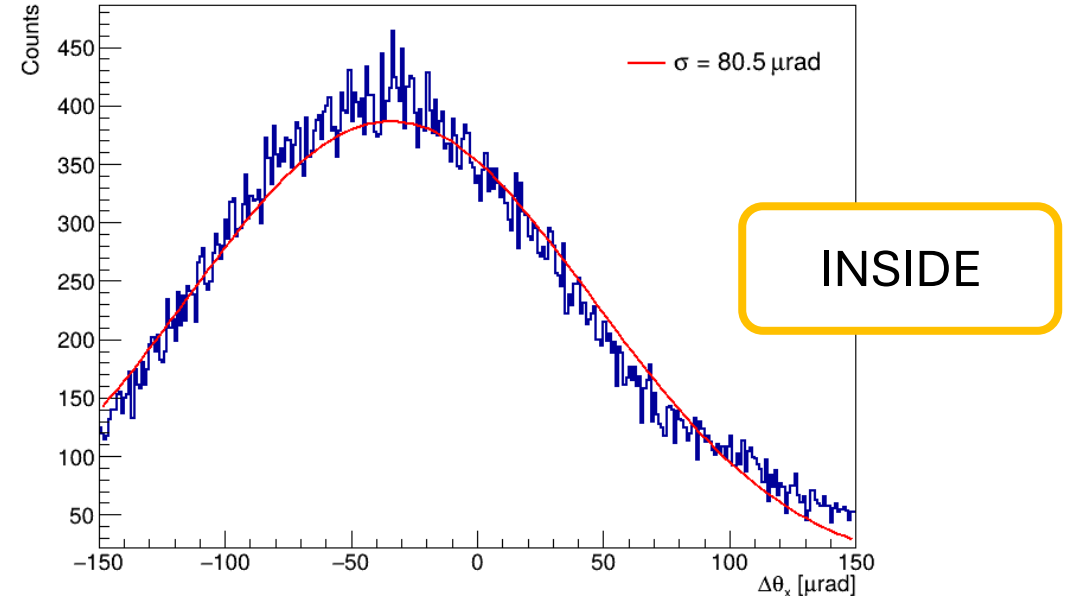
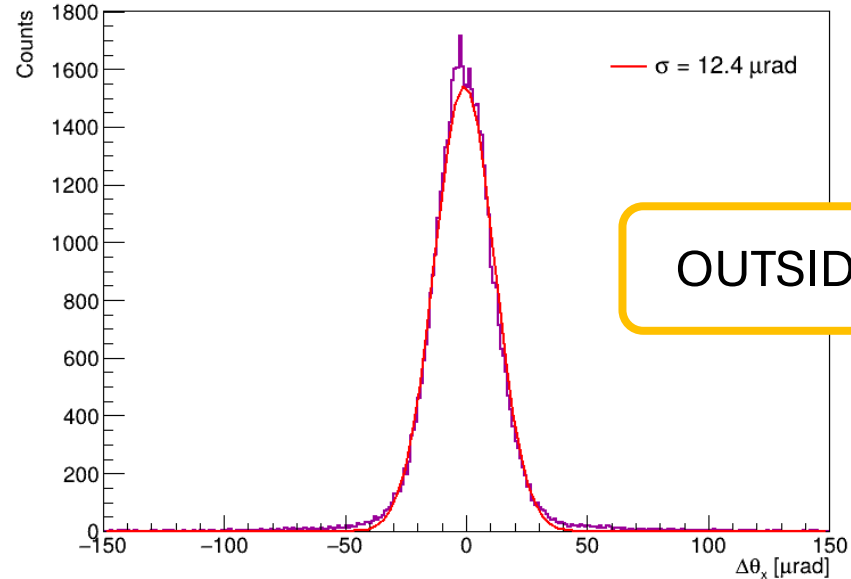
crystal 1	$(19.6 \pm 0.1 \text{ [stat]} \pm 1.5 \text{ [syst]}) \%$
crystal 2	$(18.2 \pm 0.1 \text{ [stat]} \pm 1.4 \text{ [syst]}) \%$

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

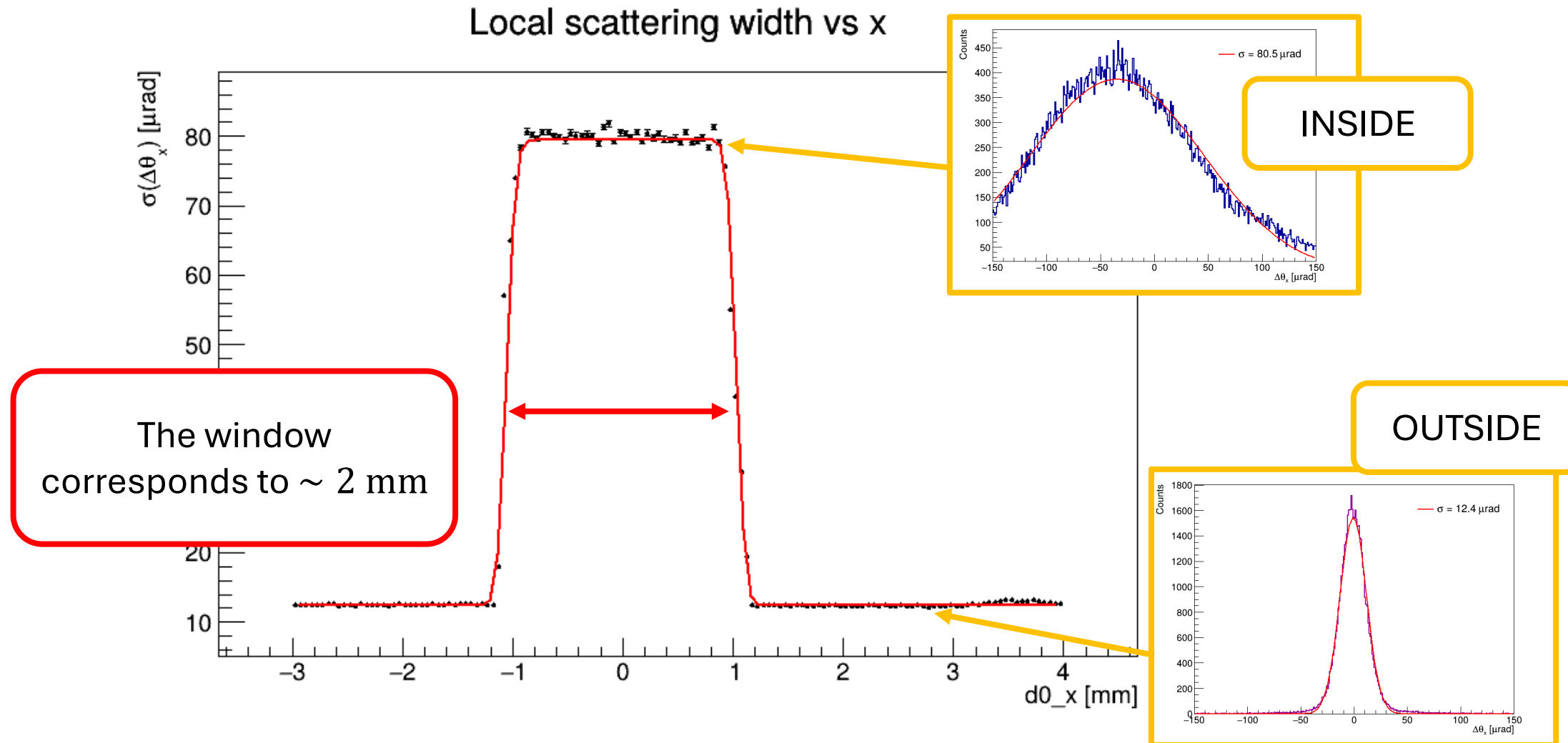
- Limitation of the previous method:
 - Isolating channeled particles to map the crystal footprint only uses a fraction of the beam ($\sim 20\%$)
 - Can introduce biases near the edges
- Using MCS:
 - Every particle that traverses the crystal interacts with its atomic lattice, picking up extra angular deflection due to MCS
 - Using 100% of the available beam statistics

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

- For each bin the angular distribution of the outgoing tracks $\Delta\theta_x$ is fitted with a Gaussian curve
- The extracted gaussian widths σ are plotted against the impact position
- The transition between the outside and inside regions is fitted using a Error function (Erf)



FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

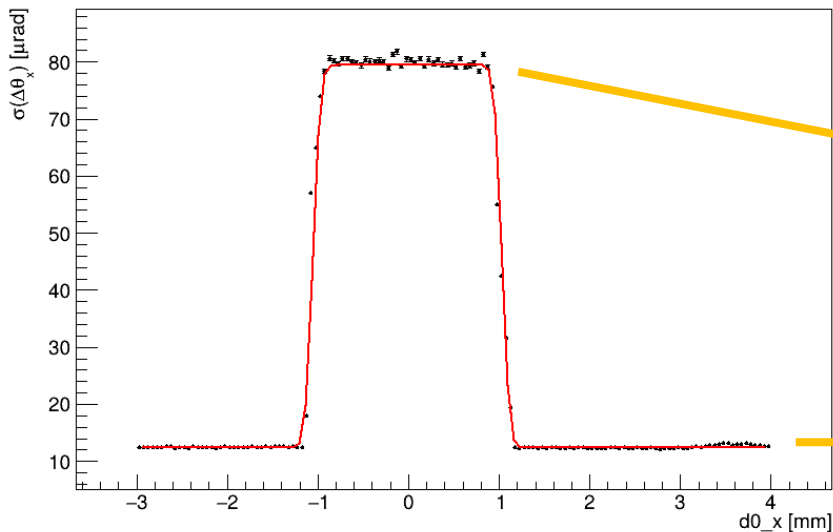


FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

$$\theta_0 = \frac{13.6 \text{ MeV}}{\beta c p} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x z^2}{X_0 \beta^2} \right) \right] = 66.5 \mu\text{rad}$$

The measured angular width in each bin is not purely the scattering form the crystal, but has the detector resolution contribution:

$$\sigma_{meas} = \sqrt{\sigma_{mcs}^2 + \sigma_{track}^2}$$



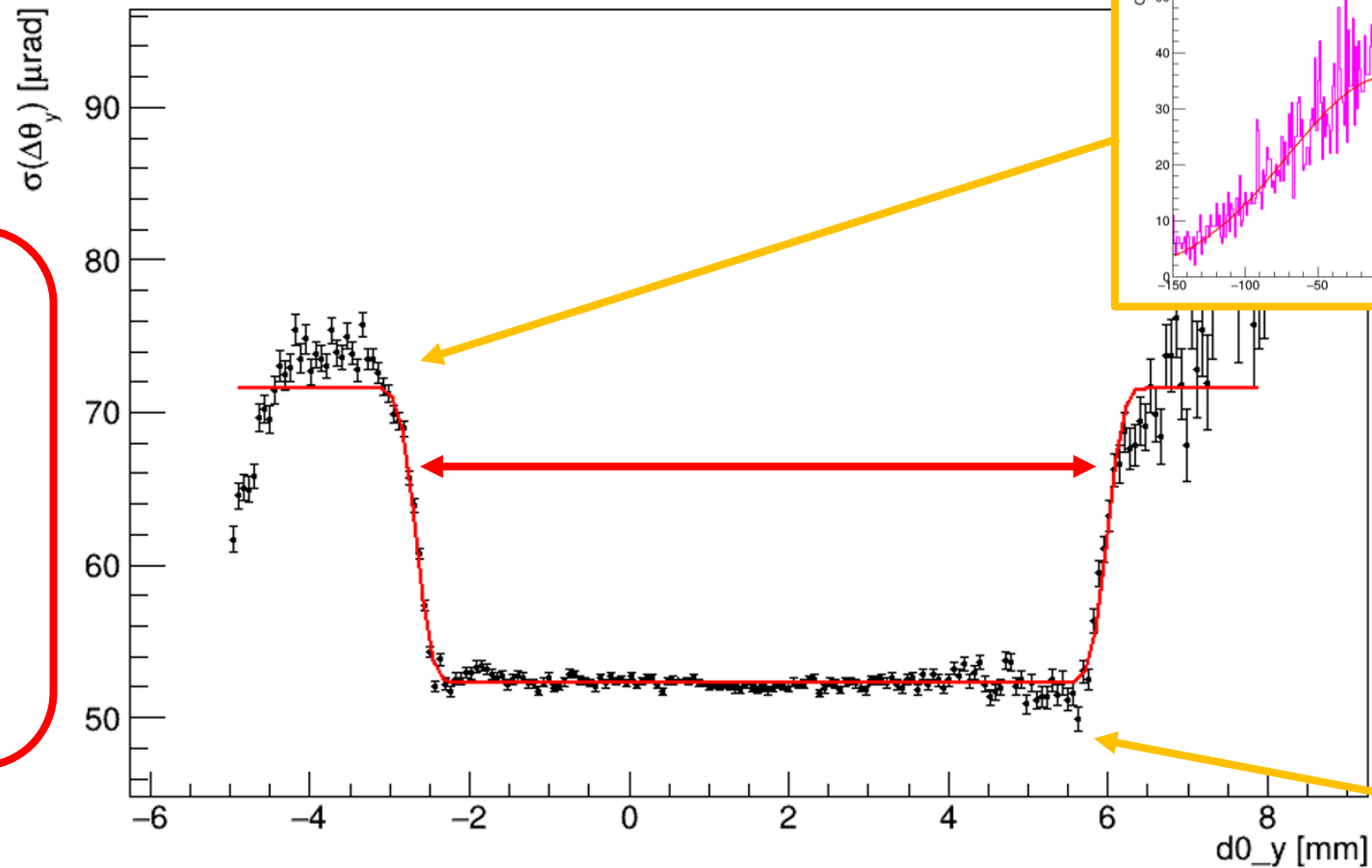
$\sim 79.5 \mu\text{rad}$

not zero, but $\sim 12.5 \mu\text{rad}$
which corresponds to σ_{track}

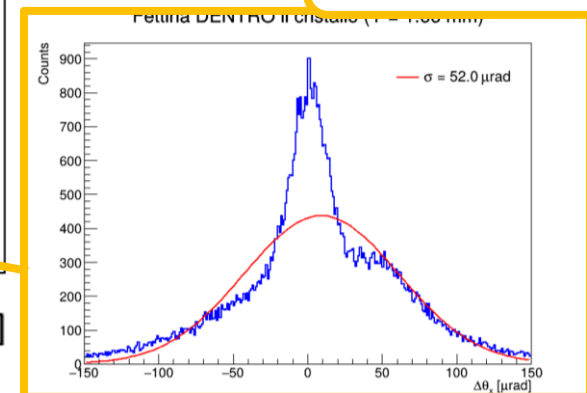
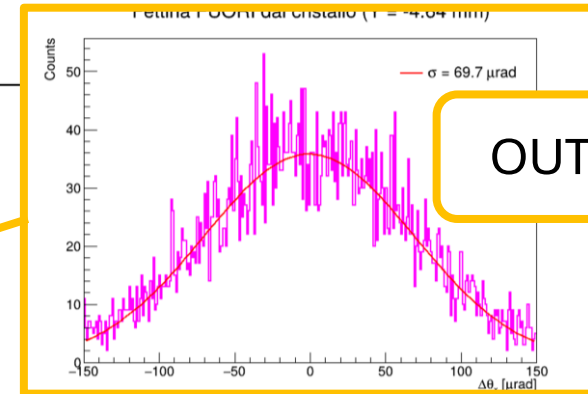
$$\sigma_{track} = \sqrt{\sigma_{track,in}^2 + \sigma_{track,out}^2}$$

FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

Local scattering width vs y



The window corresponds to ~ 8 mm while crystal measures 12.8 mm



BACKUP SLIDES
